RATIOS AND COMMON-SIZE ANALYSIS

Learning Outcome

- calculate and interpret free cash flow to the firm, free cash flow to equity, and performance and coverage cash flow ratios

In common-size analysis of a company's income statement, each income and expense line item is expressed as a percentage of net revenues (net sales). For the common-size balance sheet, each asset, liability, and equity line item is expressed as a percentage of total assets. The common-size cash flow statement has two alternative approaches. The first approach is to express each line item of cash inflow (outflow) as a percentage of total inflows (outflows) of cash, and the second approach is to express each line item as a percentage of net revenue. The common-size format makes it easier to see trends in cash flow rather than just looking at the total amount.

Consider the statement of cash flows for Acme Corporation in Exhibit 3. Exhibit 4 demonstrates the total cash inflows/total cash outflows common-size method for Acme Corporation. Under this approach, each of the cash inflows is expressed as a percentage of the total cash inflows, whereas each of the cash outflows is expressed as a percentage of the total cash outflows. In Panel A, Acme's common-size statement is based on a cash flow statement using the direct method of presenting operating cash flows. Operating cash inflows and outflows are separately presented on the cash flow statement, and therefore, the common-size cash flow statement shows each of these operating inflows (outflows) as a percentage of total inflows (outflows).

In Panel B of Exhibit 4, Acme's common-size statement is based on a cash flow statement using the indirect method of presenting operating cash flows. When a cash flow statement has been presented using the indirect method, operating cash inflows and outflows are not separately presented; therefore, the common-size cash flow statement shows only the net operating cash flow (net cash provided by or used in operating activities) as a percentage of total inflows or outflows, depending on whether the net amount was a cash inflow or outflow. Because Acme's net operating cash flow is positive, it is shown as a percentage of total inflows.

Exhibit 3:

Acme Corporation Cash Flow Statement (Direct Method) for Year Ended 31 December 2018

Cash flow from operating activities:	
Cash received from customers	$23,543
Cash paid to suppliers	(11,900)
Cash paid to employees	(4,113)
Cash paid for other operating expenses	(3,532)
Cash paid for interest	(258)
Cash paid for income tax	(1,134)
Net cash provided by operating activities	2,606
Cash flow from investing activities:	
Cash received from sale of equipment	762
Cash paid for purchase of equipment	(1,300)
Net cash used for investing activities	(538)

Cash paid for dividends	(1,120)
Net cash used for financing activities	(2,120)
Net increase (decrease) in cash	(152)
Cash balance, 31 December 2017	1,163
Cash balance, 31 December 2018	1,011

Exhibit 4:

Acme Corporation Common-Size Cash Flow Statement: Percentage of Inflows/Outflows Approach

Panel A. Direct Format for Cash Flow		
Inflows		**Percentage of Total Inflows**
Receipts from customers	USD23,543	96.86%
Sale of equipment	762	3.14
Total	USD24,305	100.00%
Outflows		**Percentage of Total Outflows**
Payments to suppliers	USD11,900	48.66%
Payments to employees	4,113	16.82
Payments for other operating expenses	3,532	14.44
Payments for interest	258	1.05
Payments for income tax	1,134	4.64
Purchase of equipment	1,300	5.32
Retirement of long-term debt	500	2.04
Retirement of common stock	600	2.45
Dividend payments	1,120	4.58
Total	USD24,457	100.00%
Net increase (decrease) in cash	(USD152)	

Panel B. Indirect Format for Cash Flow		
Inflows		**Percentage of Total Inflows**
Net cash provided by operating activities	USD2,606	77.38%
Sale of equipment	762	22.62
Total	USD3,368	100.00%
Outflows		**Percentage of Total Outflows**
Purchase of equipment	USD1,300	36.93%
Retirement of long-term debt	500	14.20
Retirement of common stock	600	17.05
Dividend payments	1,120	31.82
Total	USD3,520	100.00%
Net increase (decrease) in cash	(USD152)	

Exhibit 5 demonstrates the second method of common-sizing the statement of cash flows: the net revenue approach. Under the net revenue approach, each line item in the cash flow statement is shown as a percentage of net revenue. The common-size statement in Exhibit 5 has been developed based on Acme's cash flow statement using the indirect method for operating cash flows and using net revenue (cash received from customers) for the company in 2018 of USD23,543 from Exhibit 3.

This method is also useful to the analyst in forecasting future cash flows because individual items in the common-size statement (e.g., depreciation, fixed capital

revenue.

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Exhibit 5:

Acme Corporation Common-Size Cash Flow Statement: Net Revenue Approach

		Percentage of Net Revenue
Cash flow from operating activities:		
Net income	USD2,210	9.37%
Depreciation expense	1,052	4.46
Gain on sale of equipment	(205)	(0.87)
Increase in accounts receivable	(55)	(0.23)
Increase in inventory	(707)	(3.00)
Decrease in prepaid expenses	23	0.10
Increase in accounts payable	263	1.11
Increase in salary and wage payable	10	0.04
Decrease in interest payable	(12)	(0.05)
Increase in income tax payable	5	0.02
Increase in other accrued liabilities	22	0.09
Net cash provided by operating activities	USD2,606	11.04%
Cash flow from investing activities:		
Cash received from sale of equipment	USD762	3.23%
Cash paid for purchase of equipment	(1,300)	(5.51)
Net cash used for investing activities	USD(538)	(2.28)%
Cash flow from financing activities:		
Cash paid to retire long-term debt	USD(500)	(2.12)%
Cash paid to retire common stock	(600)	(2.54)
Cash paid for dividends	(1,120)	(4.75)
Net cash used for financing activities	USD(2,220)	(9.41)%
Net decrease in cash	USD(152)	(0.64)%

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EXAMPLE 2

Analysis of a Common-Size Cash Flow Statement

1. Andrew Potter is examining an abbreviated common-size cash flow statement for Apple Inc., a multinational technology company. The common-size cash flow statement, presented in Exhibit 6, was prepared by dividing each line item by total net sales for the same year.

Exhibit 6:

Apple Inc. Common-Size Statements of Cash Flows as Percentage of Total Net Sales

	12 Months Ended		
	30 September 2017	24 September 2016	26 September 2015
Statement of Cash Flows [Abstract]			
Operating activities:			

Net income	21.1%	21.2%	22.8%
Adjustments to reconcile net income to cash generated by operating activities:			
Depreciation and amortization	4.4%	4.9%	4.8%
Share-based compensation expense	2.1%	2.0%	1.5%
Deferred income tax expense	2.6%	2.3%	0.6%
Other	–0.1%	0.2%	0.2%
Changes in operating assets and liabilities:			
Accounts receivable, net	–0.9%	0.2%	0.2%
Inventories	–1.2%	0.1%	–0.1%
Vendor non-trade receivables	–1.9%	0.0%	–1.6%
Other current and non-current assets	–2.3%	0.5%	–0.1%
Accounts payable	4.2%	0.9%	2.1%
Deferred revenue	–0.3%	–0.7%	0.4%
Other current and non-current liabilities	–0.1%	–0.9%	3.9%
Cash generated by operating activities	27.7%	30.5%	34.8%
Investing activities:			
Purchases of marketable securities	–69.6%	–66.0%	–71.2%
Proceeds from maturities of marketable securities	13.9%	9.9%	6.2%
Proceeds from sales of marketable securities	41.3%	42.0%	46.0%
Payments made in connection with business acquisitions, net	–0.1%	–0.1%	–0.1%
Payments for acquisition of property, plant and equipment	–5.4%	–5.9%	–4.8%
Payments for acquisition of intangible assets	–0.2%	–0.4%	–0.1%
Payments for strategic investments, net	–0.2%	–0.6%	0.0%
Other	0.1%	–0.1%	0.0%
Cash used in investing activities	–20.3%	–21.3%	–24.1%

Financing activities:			
Proceeds from issuance of common stock	0.2%	0.2%	0.2%
Excess tax benefits from equity awards	0.3%	0.2%	0.3%
Payments for taxes related to net share settlement of equity awards	–0.8%	–0.7%	–0.6%
Payments for dividends and dividend equivalents	–5.6%	–5.6%	–4.9%
Repurchases of common stock	–14.4%	–13.8%	–15.1%
Proceeds from issuance of term debt, net	12.5%	11.6%	—
Repayments of term debt	–1.5%	–1.2%	0.0%
Change in commercial paper, net	1.7%	–0.2%	0.9%
Cash used in financing activities	–7.6%	–9.5%	–7.6%
Increase/(Decrease) in cash and cash equivalents	–0.1%	–0.3%	3.1%

2. Based on the information in Exhibit 6, discuss the trends in Apple's:

A. depreciation and amortization expense.

B. capital expenditures.

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less than 5 percent of total net revenue in 2015 and 2016, declining to 4.4 percent in 2017.

B. Apple's level of capital expenditures is greater than its depreciation and amortization in 2016 and 2017, whereas it was at about the same level as depreciation and amortization in 2015. In 2017, capital expenditures approached 6 percent. This is an indication that Apple is doing more than replacing property, plant, and equipment, and is expanding those investments. With cash generated from operating activities exceeding 27 percent of sales in every year, however, Apple has more than enough cash flow from operations to fund these expenditures.

3. Compare Apple's operating cash flow as a percentage of revenue with Apple's net profit margin.

Hide Solution

Solution:

Apple's operating cash flow as a percentage of sales is much higher than net profit margin in every year. This gap appears to be declining however over the three-year period. In 2015 net profit margin was 22.8 percent, while operating cash flow as a percentage of sales was 34.8 percent. By 2017, the net profit margin declined slightly to 21.1 percent, while the operating cash flow as a percentage of sales declined more to 27.7 percent. The primary difference appears to have been an increase in the level of receivables and inventory purchases, somewhat offset by an increase in accounts payable.

4. Discuss Apple's use of its positive operating cash flow.

Hide Solution

Solution:

Apple generated a large amount of operating cash flow each year, exceeding net income. This cash flow is used for relatively modest purchases of property, plant, and equipment, substantial purchases of marketable securities (investments), dividend payments and repurchases of its own stock.

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